

Abstract

Alzheimer's Disease (AD), currently the most common cause of dementia, is characterized by the accumulation of amyloid plaques and neurofibrillary tangles in the brain, leading to neuronal degeneration and brain atrophy. Patients with AD experience progressive cognitive and behavioral impairments, including memory loss, decreased concentration, impaired reasoning, and difficulties with daily functioning. Globally, over 55 million people are living with dementia, with AD accounting for 60-70% of cases. This places a substantial burden on caregivers, who often experience emotional stress, depression, anxiety, and a reduced quality of life (QoL). To address this strain, this study explores the development of an assistive ecosystem for Alzheimer's care: Adaptive Learning Zone Using Artificial Intelligence (ALZUAI). ALZUAI integrates voice assistants, motion sensors, and environmental cues within a privacy-preserving architecture, providing real-time task reminders, safety alerts, and gentle guidance, all without requiring wearable devices within the home, a known barrier in AD care. Many existing smart-home dementia studies focus on issue detection rather than active assistance, something ALZUAI would provide through proactive contextual prompts. Machine learning personalizes reminders and notifications based on behavioral trends, and caregivers receive updates via SMS or mobile application. By offering continuous, unobtrusive monitoring and proactive support, ALZUAI aims to reduce caregiver oversight burden, enhance patient safety and improve overall QoL for both patient and caregiver.

Introduction

Approximately 7.4 million Americans aged 65 and older are living with Alzheimer's Disease (AD), typically surviving four to eight years post-diagnosis. Because AD demands highly individualized support, care is largely provided by unpaid family members and friends within the community. This care requires immense flexibility, particularly when navigating the drastic behavioral changes common to AD: wandering, aggression, or inappropriate sexual behavior. Managing these unpredictable environments places a severe psychological and physical toll on caregivers. Their daily responsibilities span a wide and exhausting range, from basic tasks like grocery shopping and bill paying to intensive hands-on care like bathing and medication management (CDC and Alzheimer's Association). Consequently, despite a recent increase in available caregiver resources, full-time caregivers experience a noticeable reduction in their overall Quality of Life (QoL). The chronic stress of Alzheimer's caregiving frequently manifests as social isolation, loneliness, compromised immune function, and a higher reliance on psychoactive medications.

Over the years, several measures have been explored in order to bridge the gap between effective AD care and caregiver QoL. The introduction of smart home technologies, such as ambient and motion sensors, enable continuous, real-time tracking of behaviors like sleep patterns and gait to assist Alzheimer's patients aging in place. While these technologies improve safety and reduce caregiver stress, their effectiveness is

hindered by technical unreliability, ethical concerns regarding surveillance, and a lack of standardized evaluation frameworks (Shaik et al., 2025). To combat existing hurdles in current research, a shift from simple issue detection to proactive assistance is needed. To address this paradigm shift, this paper presents the Adaptive Learning Zone Using Artificial Intelligence (ALZUAI). ALZUAI bridges the gap between passive monitoring and active care by introducing a low-cost, multi-modal environmental framework that delivers personalized, real-time guidance without compromising user privacy and data autonomy.

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